The Criminogenic Consequence of Export Slowdown: Evidence from Millions of Court Judgment Documents in China*

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Abstract

This paper highlights the criminogenic consequence of the remarkable slowdown in China's exports in recent years. By applying textual analysis to millions of judgment documents from all levels of courts in China, we construct measures of crime rates that vary across cities over time. Our estimations, using a shift-share instrumental variable, find a higher increase in crime rates in cities that experience a more severe export slowdown. The effects are more pronounced in the manufacturing-specializing regions, where there is a higher concentration of young people, migrants, and school dropouts. Negative export shocks also lead to reduced job opportunities, decreased labor earnings for migrants, and worsened psychological health. Alternative mechanisms, such as spending on social stability, appear to play a minor role. A back-of-the-envelope calculation based on our baseline estimation suggests that approximately 13.1% of the interquartile difference in crime rates across cities can be explained by differences in their exposure to the export slowdown.

Keywords: Export slowdown; crime; courts; labor market conditions; shift-share instruments

JEL Code: F10, F14, F16, K42

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1 Introduction

Since the 2008 global financial crisis, global trade growth has been sluggish (World Bank, 2020), raising concerns about insufficient job creation, particularly in the manufacturing sector (Gomis et al., 2020).[1] This stagnation is exacerbated by a recent wave of deglobalization, marked by increasing protectionism (Fajgelbaum et al., 2020), trade wars (Amiti et al., 2019, 2020), and significant shifts in global supply chains (Alfaro and Chor, 2023; Handley et al., 2023; Antràs and De Gortari, 2020). While an extensive body of literature has explored the effects of trade shocks—such as export expansion and import competition—on income, employment, and other economic outcomes, the impacts of an export slowdown remain largely unexamined.

This paper draws on evidence from China, an economy largely export-oriented, to highlight a lesser-known socioeconomic consequence of export slowdown: the rising incidence of crime. Two features distinguish our paper from the pioneering work in this field, notably Dell et al. (2019) and Dix-Carneiro et al. (2018). First, we focus on China, which, with the world's largest working-age population, has long relied on global demand to fuel growth and job creation (Feenstra and Hong, 2010). This focus also offers a rare opportunity to examine the determinants of crime within the broader Asian context, which has been understudied due to limited data availability (He and Marshall, 1997; Yu and Zhang, 1999; Xu, 2018; Dong et al., 2020). Second, we focus on export slowdowns, which raise significant concerns in an era of deglobalization, contrasting with the predominant focus of previous studies on import competition. Over the past decade, China's export growth has declined sharply from an average of 25% annually during 2001-2008 to less than 6% annually from 2011-2018. Such a dramatic decline would inevitably result in profound socioeconomic impacts. By exploiting the vast variations in crime rates and trade exposure across Chinese cities, we provide the first comprehensive analysis of the criminogenic consequences of the pronounced deceleration in China's export growth.

The core of our empirical analysis is based on novel measures of various types of crime constructed from detailed textual analysis of judicial documents. In 2013, China's Supreme People's Court (henceforth SPC) required that all judgments from all levels of Chinese courts be made available to the public, allowing for a precise measure of crime rates across prefecture cities and over time. Furthermore, cities differ in their exposure to export shocks because

[1] According to Aslam et al. (2016), after recovering from the low of the Great Recession, the average annual growth rate of global trade is only 3.0% percent since 2012, in contrast to the pre-crisis (1987-2007) average of 7.1%. The World Employment and Social Outlook (Gomis et al., 2020) highlights that the global unemployment rate was at 5.4% in 2019 and is projected to remain at this high level in the coming years. The report also documents that the underlying reduction in employment growth is closely related to a slowdown in global economic activity, especially in the manufacturing sector.

they have different shares of the manufacturing sector; within manufacturing, they tend to specialize in different products. Therefore, we measure each city's export exposure using a shift-share à la Bartik approach and employ a shift-share instrumental variable (IV) to identify the causal effect. The validity of our approach relies on the construction of foreign shocks that reflect aggregate demand changes in the export destinations but are independent of Chinese firms' export decisions. To proxy demand shifts for Chinese products, we use the change in total imports of twenty-eight OECD countries in Europe and America from nations with export structures most similar to China's. The exposure of each city is then measured by averaging global import shifts by product, weighted by the composition of city exports by product in 2010. This allows us to compare changes in crime rates in cities that were more and less intensely affected by the declining demand for Chinese exports.

We document robust evidence that China's export slowdown led to rising crime rates. According to our preferred two-stage least squares (2SLS) estimation, a reduction of $1,000 in a city's export per worker is predicted to induce 143 additional criminal cases per million population. The increase in crime rates is sizable, considering that the average in our sample is 447 cases per million population. A back-of-the-envelope calculation based on our baseline estimation suggests that approximately 13.1% of the interquartile difference in crime rates across cities could be explained by the difference in their exposure to the export slowdown.[2] As crime has an important externality dimension, our findings call for more policy attention to the socioeconomic consequences of trade-induced shocks.

These empirical findings are robust to a variety of robustness and falsification tests as suggested by recent shift-share literature (Adao et al., 2019; Borusyak et al., 2022; Goldsmith-Pinkham et al., 2020). We demonstrate the absence of differential pre-trends in crime rates by investigating the lag and lead effects of export slowdowns. Our benchmark estimates are not driven by a few large provinces or a small number of products, nor do they depend on initial specialization patterns or differential crime rate trends across cities. They are also robust to various domestic shocks, spatial correlation in export shocks across cities, and clustering of standard errors at alternative levels. The consistency in the sizes and patterns of the estimates bolsters our confidence that the export slowdown leads to increased crime rates across cities.

We also observe significant variations in the criminal response to adverse export shocks based on city characteristics and types of crime. Cities that rely heavily on manufacturing and have higher proportions of young people, migrants, or school dropouts are more adversely

[2] Another way to gauge the significance of our estimate is that suppose China had maintained its pre-crisis annual export growth at 20%, the crime rate in Chinese cities would have decreased by 12.1% from its current level.

affected by the export slowdown. In terms of crime types, we find that the export slowdown has a more pronounced effect on offenses involving violence, prostitution, gambling, felony traffic collisions, and criminal activities related to resource appropriation, such as robbery and stealing.

Classic theory (Becker, 1968) suggests that shrinking job opportunities or earnings reduce the opportunity cost of committing crimes. Our findings support this insight: we estimate that a \$1,000 reduction in exports per worker leads to a 12% decline in total employment and a 8.8% drop in the average wage of migrants, with these effects particularly pronounced in the manufacturing sector. The deterioration of economic conditions can also induce stress or psychological distress among workers, which may contribute to the rising incidence of violent crimes (Frijters et al., 2012; Colantone et al., 2019). We observe that adverse export shocks significantly worsen mental health conditions for individuals employed in the manufacturing sector, as well as for young people aged 16 to 30. Cities experiencing a sharper decline in exports also see reduced inflows and increased outflows of migrant workers. As an alternative explanation, sluggish exports may reduce government spending on public security or surveillance equipment, thereby inducing criminal activities (Fishback et al., 2010; Dix-Carneiro et al., 2018; Bennett and Ouazad, 2020). However, we find that this channel plays only a minor role.

The paper contributes to several strands of literature. First, our study joins the fast-growing literature that estimates the effects of trade shocks on the local labor markets. Following the influential study of Autor et al. (2013), many studies have examined the effects of import competition from low-wage countries, particularly China, on local employment and wages.[3] We join the growing attention on the impact of export shocks on the local economy (Dauth et al., 2014; Feenstra et al., 2019; Erten and Leight, 2021; etc.). We share the rising concern about how the export shocks affect the local economy. Our paper is among the first to examine the rising social tensions, following negative export shocks. A closely related study in this regard is Campante et al. (2023), which documents rising labor strikes in Chinese prefectures because of the export slowdown.

We speak to the literature that connects trade and crime. On the one hand, trade liberalization disincentivizes illegal trade and, therefore, reduces crime incidents (Prasad, 2012). On the other hand, rising import competition is found to destroy jobs and lower wage earnings, which may induce more crime incidents as the opportunity cost of being a criminal is lowered (Dell et al., 2019; Iyer and Topalova, 2014; Che et al., 2018; Dix-Carneiro et al.,

[3] Related studies include Dix-Carneiro and Kovak (2015), Balsvik et al. (2015), Acemoglu et al. (2016), Feler and Senses (2017), and Dix-Carneiro and Kovak (2017) among others. Autor et al. (2016) provide an excellent survey on the impact of China's import shock.

2018).[4] Our focus on rising crime rates in Chinese cities due to shrinking export opportunities has been rarely explored in previous work. We find that export slowdown induces more crime, mainly through weakening local labor market conditions, which is consistent with the Brazilian evidence documented in Dix-Carneiro et al. (2018).

Our study, building on the pioneering Becker-Ehrlich models (Becker, 1968; Ehrlich, 1973), delves into the economic motives behind criminal activities. Previous research suggests declining employment and income opportunities lower the opportunity cost of committing crimes, thus increasing criminal behavior (Becker, 1968; Entorf and Spengler, 2000; Raphael and Winter-Ebmer, 2001; Gould et al., 2002; Machin and Meghir, 2004; Lin, 2008; Fougère et al., 2009; Gronqvist, 2013; Rose, 2018; Bennett and Ouazad, 2020; Axbard et al., 2021; Britto et al., 2022), while access to credit, public transfer, and relief spending are found to attenuate criminal activities (Khanna et al., 2021; Che et al., 2018; Fishback et al., 2010; Britto et al., 2022).[5] Consistent with these findings, we observe that the export slowdown significantly harms the local economy and labor market outcomes. Additionally, local governments tend to increase spending on social stability in response to the decline in exports.

The final contribution of our study is to construct a city-level measure of crime rates for various criminal activities. This is not the first study to examine the determinants of crime rates in China. However, previous studies have mainly relied on the province-level published statistics, as reported in the China Procuratorial Yearbook. For instance, Edlund et al. (2013) uses a province-level panel to study how the sex ratio imbalance led to the high incidence of violence and property crime in China between 1988 and 2004. Li et al. (2019) explores the relationship between income inequality and criminal prosecution rates. These official statistics may be underreported (He and Marshall, 1997; Yu and Zhang, 1999; Dong et al., 2020) and do not distinguish between specific types of crime. By contrast, our crime statistics are directly derived from the court judgment documents, which are legally published and publicly accessible. Specifically, we adopt a new textual analysis approach to analyze the effective judgments, verdicts, and conciliation agreements at all levels of Chinese courts, which help measure the crime rate for each Chinese city and by fourteen types of criminal activities.[6] With this new set of crime measurements, we offer further explanations for the rising crime rates across Chinese cities based on those cities' exposure to the recent export slowdown.

[4]In addition, Garfinkel et al. (2008, 2015) find that trade liberalization may induce conflicts when the enforcement of property rights is imperfect, and trade intensifies competition for natural resources.

[5]However, Bennett and Ouazad (2020) and Bhalotra et al. (2021) find that public transfers, such as unemployment benefits, do not necessarily offset the negative impact of income loss on crime.

[6]Dong et al. (2020) use the judgments documents from 2014 to 2016 to study the impact of poverty and inequality on homicide rates in China. In comparison, we identify the causal effect of export shocks on crime. In addition, our study covers all types of crime in a longer period.

The rest of the paper is organized as follows. We describe the various data sources and statistical measures used in our analysis in section 2, and introduce the empirical strategy in section 3. Section 4 presents the main empirical findings. Section 5 discusses possible mechanisms, and section 6 concludes the paper.

2 Data Sources and Measures

We describe our data sources and main variables, including the measures of crime rates and export shocks. Variables used in mechanism analysis are discussed in section 5. Further details on the data construction are described in Appendix C. Throughout our study, we focus on prefecture cities as the geographical unit.[7] Our sample includes 325 prefectures covering 31 provinces in mainland China, with a median working-age population (15-64 year old) of 2.35 million in 2010 and a median land area of 12,874km^2.

2.1 Measuring Crime Rates

Our crime data are obtained from the China Judgments Online Database (CJO), the official website of the SPC of China. As a crucial component of a judicial reform aimed at increasing legal transparency, the SPC mandates that all levels of people's courts disclose their judicial documents on an online platform.[8] The CJO database, therefore, includes nearly all types of judgment documents from 2012 to 2018, except for certain special cases such as those involving national security, juvenile delinquency, divorce litigation, and cases resolved through mediation. The CJO database comprises five main types of documents: criminal, civil, administrative judgments, enforcement rulings, and other judicial documents (such as payment orders, state compensation decisions, and mediation documents). Each judgment document includes the prosecution year, title, court location, and a detailed description of the event. In this study, we focus on criminal cases that violate the Criminal Law of the People's Republic of China due to the high social costs associated with criminal

[7]China's administrative divisions follow a four-layer hierarchy. At the top are provincial units, including provinces (e.g., Jiangsu), autonomous regions (e.g., Tibet), and municipalities directly under the central government (e.g., Beijing, Shanghai, Chongqing, Tianjin). The second level is the prefecture, which is further divided into (autonomous) counties and county-level cities. The fourth level consists of townships or towns. This study focuses on prefecture-level cities. Our analysis includes the four municipalities and provides robustness checks excluding them. For details, see `http://xzqh.mca.gov.cn/statistics/2018.html`.

[8]People's courts are the judicial organs of China responsible for case trials and dispute settlement. The court system in China is organized into four levels: over 3,000 county-level primary courts, over 400 city-level intermediate courts, 31 province-level high courts, and one supreme court. Each level of court can serve as a court of first instance, but they handle different types of criminal cases. Higher people's courts have the authority and duty to supervise the work of lower people's courts. For more detailed information on China's legal system, see Appendix B. The CJO was established on November 13, 2013. Detailed information about the judgment documents covered by the CJO can be found in Appendix C. For additional details, please visit the official website at `https://wenshu.court.gov.cn/`.

activities (Loayza et al., 2000), and refine the sample to judgment documents with complete information on the occurrence time and location of the criminal activity.[9] For each case, we extract the actual time when criminal activities occurred from the case description transcript and use all first-instance criminal cases sentenced by all levels of courts in a city to calculate the city-level crime rate, as most criminal cases are heard and sentenced by county-level primary courts or city-level intermediate courts.[10]

Variable construction: We conduct a textual analysis of case titles and description transcripts to categorize each case in our data by type of crime. Specifically, we search each judgment document for keywords related to specific crimes. For example, a case is classified as "Robbery" if it contains the Chinese keywords "抢劫," "抢夺," or "哄抢," which all translate to "robbery" in English. For cases involving several types of criminal activities, we assign them to multiple categories. For instance, if a criminal committed both robbery and homicide simultaneously, we assign this case to both "Robbery" and "Violence". Appendix Table A.1 summarizes the keywords associated with the fourteen types of crime studied in this paper. Further, we aggregate the individual-level data to obtain the total number of crime incidents by year, city, and crime type. We standardize the number of criminal cases by the population with residency rights (Hukou) of each city in 2010, based on the 2010 Population Census, to measure crime rates because cities differ greatly in size. Thus, our dependent variable measures the number of criminal cases per thousand population.[11]

Table 1 shows the summary statistics, where we report the mean and the standard deviation (in parentheses) of city-level crime rates by region (i.e., East, Middle, and West China) and three time bins from 2011 to 2018, with the last column applying to all sample years. Overall, the eastern region has higher crime rates than the western and central China. And it also experiences the most significant increase in crime rates, with the average crime rate rising from 0.25 (2011-2013) to 0.59 (2017-2018). Figure 1 illustrates the spatial

[9]According to Miller et al. (1993) and Levitt (2002), crime costs victims \$200 billion per year in the United States, while the cost of reducing it is equally high. In Latin America, the estimated social costs of crime amount to \$168 billion, or 14.2 percent of the region's GDP (Loayza et al., 2000). As for China, the rapid growth of criminal activities in recent years has attracted a lot of attention (Edlund et al., 2013; Cameron et al., 2019).

[10]In our sample, only a small share (3.78%) of the case documents lack precise information on the time and location of the incidents. The primary reason for missing information is irregular statements or simply missing details about the actual time of criminal activity or the location of the court in the raw data. In addition, there are also cases for which the descriptions of the crime and the subsequent judgment are complex, making it difficult for natural language processing (NLP) procedures to accurately extract the information. Criminal cases closed by the province-level high courts and the SPC only account for 0.003% (116/3,719,557) in our sample.

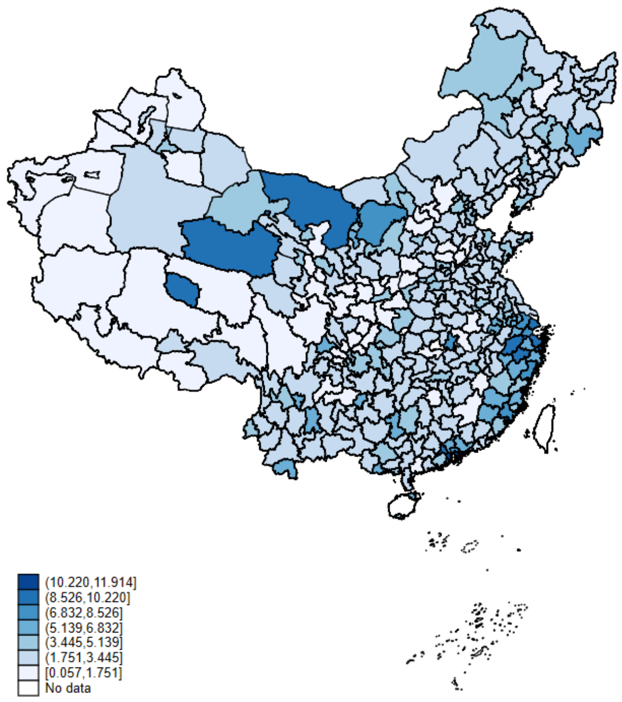
[11]In our empirical analysis, the main findings remain robust even when we normalize the number of criminal cases by the working-age population (ages 15-64) of each city in 2010, encompassing all individuals, both with and without Hukou.

Table 1: Summary Statistics for Crime Measures

Variables: Crime Rates	Region	2011-2013	2014-2016	2017-2018	All Years
Number of total criminal cases per thou. population	East	0.247	0.662	0.590	0.488
		(0.279)	(0.433)	(0.367)	(0.410)
	Middle	0.151	0.452	0.461	0.341
		(0.129)	(0.185)	(0.193)	(0.224)
	West	0.147	0.476	0.456	0.348
		(0.158)	(0.313)	(0.321)	(0.310)
Number of criminal cases related to public security and the safety of individuals and property per thou. population	East	0.174	0.491	0.453	0.363
		(0.211)	(0.319)	(0.278)	(0.309)
	Middle	0.102	0.354	0.375	0.264
		(0.101)	(0.144)	(0.153)	(0.183)
	West	0.098	0.362	0.360	0.263
		(0.115)	(0.252)	(0.271)	(0.251)
Number of criminal cases related to drugs, prostitution and gambling per thou. population	East	0.033	0.114	0.073	0.073
		(0.059)	(0.124)	(0.066)	(0.097)
	Middle	0.014	0.057	0.051	0.039
		(0.022)	(0.047)	(0.040)	(0.042)
	West	0.018	0.074	0.063	0.050
		(0.033)	(0.078)	(0.070)	(0.068)
Number of criminal cases related to disrupting the order of the socialist market economy per thous. population	East	0.016	0.016	0.007	0.014
		(0.012)	(0.017)	(0.008)	(0.014)
	Middle	0.014	0.008	0.004	0.009
		(0.009)	(0.006)	(0.004)	(0.008)
	West	0.011	0.006	0.002	0.007
		(0.009)	(0.005)	(0.003)	(0.007)

Notes: The table reports the summary statistics of crime rates by year, region and major types of crime. Crime rate is measured as the number of criminal cases per thousand population. The mean across cities and every year bin is reported, with the standard deviation in parentheses. The "All Years" column reports the summary statistics where we pool all data together. Provinces are classified into three regions: (i) East (including Beijing, Tianjin, Hebei, Liaoning, Shanghai, Jiangsu, Zhejiang, Fujian, Shandong, Guangdong, Hainan), (ii) Middle (including Shanxi, Jilin, Heilongjiang, Anhui, Jiangxi, Henan, Hubei, Hunan) and (iii) West (including Sichuan, Chongqing, Guizhou, Yunnan, Xizang, Shaanxi, Gansu, Qinghai, Ningxia, Xinjiang, Guangxi, Inner Mongolia). For space-saving, we divide the criminal cases into three main categories: (i) crime related to public security and the safety of individuals and property (which include public security, traffic, violence, robbery, stealing, defraud, extortion cases); (ii) crime related to drugs, prostitution and gambling; (iii) crime related to disrupting the order of the socialist market economy (which include fake-product, IP infringements, finance, bribery cases). In the following heterogeneity analysis, we further divide the criminal cases into fourteen categories.

Figure 1: Geographic Distribution of Court Recorded Crime



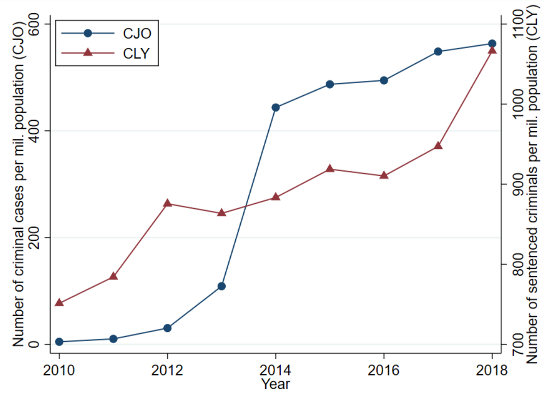
Notes: The figure displays the distribution of crime rates, measured by the number of criminal cases per thousand population, between 2012 and 2018 across China. We construct crime rates using micro-data from the China Judgments Online.

variations in crime rates across cities. The Yangtze River Delta, the southeast coastal areas, and a few western cities in Inner Mongolia and Qinghai have higher crime rates.

Data representativeness: Despite the broad geographic coverage, there are concerns about the representativeness of our measurements of crime rates, such as the variations in the crime rates over time can be driven by the courts' increasing tendency to disclose judgment documents, rather than capturing the actual increase in criminal activities. As shown in Table 1, the number of criminal cases sharply increased in the 2014-2016 sample. This is because the CJO was not established until the end of 2013; therefore, not all judgments closed before 2014 were posted online. However, this gap is unlikely to impact our estimates. As we show later, the results remain robust even when we exclude the early period, 2012 and 2013, from our sample. Furthermore, we calculate crime rates using alternative measures directly from the official statistics, which are only available at the provincial or national level. In particular, Figure 2 compares the crime rates calculated from the CJO data with two alternative official statistics sourced from the China Law Yearbook (CLY), with all measures aggregated at the national level. Panel (1) displays the CLY-based crime rates using the number of registered criminal cases per million population, while panel (2) shows

Figure 2: Comparison of Crime Data between CJO and CLY

(1) Number of criminal cases



(2) Number of sentenced criminals

Notes: The figure compares national crime rates computed from the CJO database and the China Law Yearbook (CLY), respectively. In panel (1), the CLY crime rates are measured by the number of registered criminal cases in the first trial per million population, and panel (2) uses the number of sentenced criminals per million population. In both panels, the CJO-based crime rates are measured by the number of filed criminal cases per million population.

the number of sentenced criminals per million population. In both panels, the CJO-based crime rates are measured by the number of filed crime cases per million population, ensuring consistency with the CLY data. The four series evolve closely.[12] Notably, the calculated crime rates are lower than those from the CLY annual reports in the early years, reflecting the possible under-reporting of criminal cases in these years. Importantly, the discrepancy in crime rates based on the CJO data and the official statistics is unlikely to correlate with the local economic conditions, including export shocks, our interest variable.[13]

2.2 Measuring Export Slowdown

Data on exports by product and city are taken from the China Customs Database (2010-2017) collected by China's General Administration of Customs. The database covers the universe of Chinese import and export transactions. For each transaction, it provides detailed information on trade values (in US dollars) at the eight-digit Harmonized System (HS)

[12] Appendix Figure A.1 compares the CJO data with the CLY for the top three crime categories as defined by the CLY: crimes endangering public security (including offenses related to public security and traffic felonies), crimes of property violation (including offenses related to robbery, theft, defraud, and extortion), and crimes obstructing the administration of public order (including offenses related to drugs, prostitution, gambling, creating disturbances, or criminal syndicate activities). Together, these categories account for approximately 78% of all crime cases. As shown in the figure, the two series evolve closely for all three types of crime, resembling the pattern displayed in Figure 2.

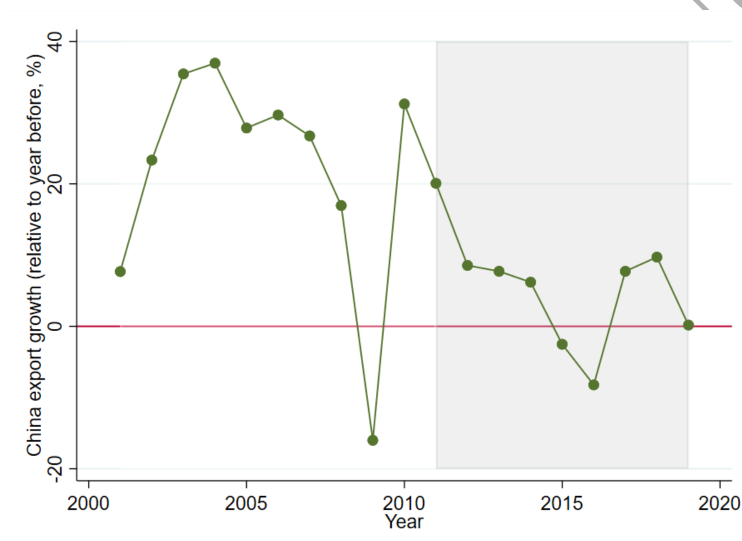
[13] Panel (1) of Appendix Figure A.2 compares our constructed crime rates based on the CJO data with the number of criminal suspects under public prosecution. The data on criminal suspects is from the China Procuratorial Yearbook and is available at the provincial level, and panel (2) of Appendix Figure A.2 compares the changes in the provincial crime rates constructed from both sources, again showing very similar patterns.

product level (over 7000 products), trade partners (i.e., destination markets and source countries), and the location (city) of the Chinese importers and exporters. To measure a city's exposure to a global trade slowdown, we follow Campante et al. (2023) and define $ExpShock_{ct}$ as the annual change in the city-level exports, standardized by the city's working-age population in 2010, the beginning of our sample period:

$$ExpShock_{ct} = \frac{Export_{ct} - Export_{ct-1}}{L_{c,2010}}, \tag{1}$$

where $Export_{ct} = \Sigma_k \Sigma_{f \in c} X_{fckt}$ is the total export value (in 1000 USD) of product k produced by firm f in city c and year t; $L_{c,2010}$ denotes the working-age population of city c in 2010, based on the 2010 China Population Census. As we focus on manufacturing exports, we concord each eight-digit HS product code to the Standard Industrial Classification (SIC) at the 4-digit level, and only retain manufacturing products whose leading digit in the SIC codes is equal to 2 or 3.[14]

Figure 3: Manufacturing Export Growth of China



Notes: The figure plots China's manufacturing exports growth rate between 2001 and 2019. The growth rate is computed as the annual changes in China's manufacturing exports to the rest of the world relative to the previous year. Exports data come from China's General Administration of Customs.

[14]The concordance between HS code and SIC product classification is provided by the World Integrated Trade Solutions (WITS) at `https://wits.worldbank.org/product\_concordance.html`. We focus on manufacturing exports for two reasons. First, manufacturing exports constitute the majority of China's export activities. In 2010, 96.8% of China's exports were manufacturing goods. This dominance has remained consistent over time, with the average share of manufacturing exports at 96.6% between 2010 and 2019. Second, since 2009, global trade has experienced a significant downturn amid rising trade barriers, particularly affecting manufactured goods (Fund, 2019). To ensure the robustness of our results, we also incorporate all industries in the construction of both the export slowdown measure and the instrumental variable. When all industries are considered, the baseline results remain stable, i.e., the estimated impact of the export slowdown on the crime rate is -0.095 with a robust standard error of 0.046.

Figure 3 plots the annual growth rates of China's manufacturing exports from 2001 to 2019. China's exports grew rapidly after its accession into the WTO in 2001, with an average annual rate of 25% between 2001 and 2008. However, the 2008-2009 global financial crisis put an end to the extraordinary growth path. Since 2011, China's export growth has declined from nearly 20% in 2011 to only 0.2% in 2019, with significantly negative growth rates in both 2015 and 2016.[15] To avoid the confounding factors owing to the China-US trade war and the pandemic, our empirical analysis focuses on the period up to 2018.

Appendix Table A.2 reports the summary statistics of the variables used in our empirical analysis. Variables in panel A are used in the baseline regressions in section 4, while variables in panel B are used in the mechanism analysis in section 5.

3 Empirical Strategy

We now describe our regression specification and the IV strategy adopted to identify the effects of export slowdown on the incidents of crime.

3.1 Baseline Specifications

We regress the change in city-level crime rate in year t on the export shock experienced by the city in year $t-1$, using the following baseline specification:

$$\triangle(Crime/L)_{ct} = \beta_1 ExpShock_{ct-1} + \beta_2 X_{ct-1} + \gamma_t + \gamma_c + \epsilon_{ct} \tag{2}$$

where $\triangle(Crime/L)_{ct} \equiv (Crime/L)_{ct} - (Crime/L)_{ct-1}$ represents the change in the number of criminal incidents per thousand population in city c between year t and $t-1$. The main explanatory variable of interest, $ExpShock_{ct-1}$, captures the exposure to export slowdown in city c and year $t-1$, constructed according to equation (1).[16] We lag export shock by one period to capture the fact that the crime rate may take time to respond to external demand shock, consistent with recent studies that emphasize the lagged effects of macroeconomic shocks on social outcomes (Caliendo et al., 2019; Perla et al., 2021). This approach allows us to capture the lagged effects of export fluctuations on crime rates, providing a more accurate and comprehensive understanding of the temporal dynamics involved. The variable X_{ct-1} is a collection of additional controls that capture other city-year-level economic or

[15] Throughout the whole period between 2011 and 2019, China only managed to maintain an average annual growth rate at 5%.

[16] According to Roth and Sant'Anna (2023), both the dependent and the main independent variables are measured on a per capita basis in regressions to ensure scale invariance and comparability, which mitigates issues related to functional form sensitivity, and reduces heteroscedasticity. This approach provides more consistent and reliable results for analysis, particularly when comparing across different cities or regions. In the baseline specification, we normalize both crime rates and export shocks using the 2010 population. However, our results remain robust when we normalize them using the contemporary population instead.

demographic factors that may influence crime rates. The same first-difference specification has been use in Dell et al. (2019) and Campante et al. (2023). In our context, a first-difference specification is preferrable to a level regression with the city fixed effects, since both the outcome variable and the main independent variable may exhibit autocorrelation (Brüderl and Ludwig, 2015).[17] The variable γ_t is the year fixed effects, which account for macroeconomic shocks or business cycles.

We also include city fixed effects (γ_c) to control for time-invariant city-specific characteristics that may affect crime rates. There are two main reasons for including the city fixed effects in a first-difference specification. First, the city fixed effects account for cross-prefecture differences in $\triangle(Crime/L)_{ct}$, or equivalently city-specific linear time trends in $(Crime/L)_{ct}$. Therefore, including such fixed effects helps to control for possible pre-trends associated with city characteristics.[18] Second, the city fixed effects also help to eliminate the possible correlation between initial export structure, which is the share in the Bartik instrument, and prefecture-specific linear trends in crime rate, as suggested by McCaig (2011) and Li (2018). The standard errors are clustered at the city level to address potential correlations within cities across different years. Our baseline estimators cover 325 cities from 2012 to 2018.

3.2 Instrument Variable

Directly estimating equation (2) using the ordinary least square (OLS) method may be problematic. First, there may be reverse causality in that higher crime rates may discourage investment and production, leading to negative export performance. Second, there could also be an omitted variable bias such that other unobserved shocks across cities and years may simultaneously affect exports and crime incidence.

To address such endogeneity, we follow Autor et al. (2013) and Campante et al. (2023) and employ a shift-share à la Bartik IV approach that does not rely on the local export performance but rather on the changes in the total imports of OECD countries from nations with export product mixes similar to China. Specifically, we identify the top countries whose export compositions closely match those of China by calculating cosine similarity scores

[17] For example, legal reporting practices typically adhere to consistent structures, which can propagate similar types of classification errors from year to year. Judicial resources, such as staffing levels or technology, also affect the depth of case documentation, resulting in a persistent under-reporting pattern across periods. Additionally, social and political pressures that evolve gradually can influence how crime is reported, often increasing focus on specific crime types in ways that yield autocorrelation errors. As for the instrumental variable, the initial export shares may consistently misrepresent local trade volumes over time. The demand-based component relies on foreign import data, which often face persistent country-specific reporting differences.

[18] For instance, if a city has a poor security environment, it is likely to experience a greater change in its crime rate.

across the HS 6-digit product categories using export data in 2010. A detailed description of this procedure is provided in Appendix D. This approach is based on the premise that countries with export profiles similar to China are likely to experience declines in global demand in a manner analogous to China, which are further less likely to be related to China's domestic local productivity shocks. Taking advantage of the differential impact of import demand from OECD countries on nations with export profiles similar to China, the new IV aims to capture exogenous variations in Chinese exports driven by shifts in external global demand.[19] The IV is constructed according to the following equation:

$$ExpShock_{ct}^{IV} = \Sigma_k \frac{X_{ck,2010}}{\Sigma_c X_{ck,2010}} \frac{\Delta X_{kt}}{L_{c,2010}}, \tag{3}$$

where $L_{c,2010}$ is the working-age population in the city c and year 2010; the variable $\Delta X_{kt} = X_{kt} - X_{kt-1}$ denotes the change in imports of each 6-digit HS product k from top nine countries whose export compositions closely match those of China to twenty-eight OECD countries in Europe and America between t and $t-1$, reflecting changes in foreign demand of Chinese products.[20] The term $X_{ck,2010}/\Sigma_c X_{ck,2010}$ denotes the start-of-period product export share, computed by dividing exports of product k from city c (i.e., $X_{ck,2010}$) by China's total export of product k (i.e., $\Sigma_c X_{ck,2010}$). In short, cities with a higher share of products experiencing a larger external export slowdown would be adversely affected more than other cities.[21]

The exogeneity of our IV relies on the assumptions that, conditional on year- and city-fixed effects, other unobserved time-varying city-specific determinants of the outcome variable in the error term are uncorrelated with: (1) the product-specific foreign demand shocks observed at the national level and (2) the initial city-product-specific export structure. Regarding the first condition, the shifts in the trade shocks (ΔX_{kt}) capture foreign demand shocks external to China. For instance, Aslam et al. (2016) provides evidence that three-fourths of the slowdown in global import growth after 2012 can be attributed to weaknesses in

[19] This strategy is conceptually similar to the instrumental variable approach used by Autor et al. (2013), who use imports by eight other developed countries from China as a proxy for U.S. demand for Chinese products.

[20] The twenty-eight OECD countries include Austria, Belgium, Canada, Czechia, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Iceland, Ireland, Italy, Luxembourg, Mexico, Netherlands, Norway, Poland, Portugal, Slovakia, Slovenia, Spain, Sweden, Switzerland, Turkiye, United Kingdom and USA. Data on product-level trade flows is from the CEPII database, which provides detailed bilateral trade flow for each six-digit HS product code. The nine nations with export structures similar to China include Czechia, Hungary, Romania, Philippines, Malaysia, Mexico, Estonia, Vietnam and Korea. As robustness checks, we also use the changes in OECD's imports from the top five or fifteen countries with the most similar export structure to China, and results remain barely changed.

[21] To ensure our results are not driven by outliers, the $ExpShock_{ct}^{IV}$ is winsorized at the 1st and 99th percentiles in the regression. The results remain stable when using alternative winsorization thresholds, such as the 0.5th and 99.5th percentiles, or the 0.1th and 99.9th percentiles.

domestic absorption and overall economic environments, particularly in advanced economies. Therefore, the variations in the Bartik IV shifts are primarily driven by changes in foreign demand conditions and are less likely to be influenced by local economic conditions. Furthermore, we conduct a series of robustness checks (in section 4.3) demonstrating that our estimates remain robust when (1) controlling for measures of city-level shocks to domestic demand, supply, and imports and (2) performing the balance test suggested by Borusyak et al. (2022), which indicates that the shifts are effectively randomly assigned to Chinese cities.

The second condition can be violated if the occurrence of criminal activities is correlated with the local initial export structure. Adding the city fixed effects (γ_c) partially eliminates this concern, as it is intended to account for any city-specific linear trends in the outcome variable that could be correlated to the initial export structure (McCaig, 2011; Li, 2018; Campante et al., 2023; Xu, 2020). In addition, we show that our estimates do not suffer from the pre-trend and pass other statistical tests suggested by Goldsmith-Pinkham et al. (2020), which we discuss in more detail in Section 4.3.

4 Empirical Results

We now discuss our main findings on the impact of the 2011-2017 export slowdown on changes in crime rates in Chinese cities from 2012 to 2018. We begin by presenting the baseline specification, followed by an analysis of heterogeneity, and discuss the robustness of our findings and provide validity checks for the Bartik instrumental variable.

4.1 Baseline Results

Table 2 presents our baseline results based on equation (2). Column (1) shows the OLS estimation, and column (2) displays the 2SLS estimation using the Bartik IV. Columns (3) and (4) include additional controls and province-year-specific trends. Columns (5) to (7) directly regress the crime rate on the instrument. The OLS result in column (1) is positive but insignificant, while the IV estimation in column (2) reveals that an export slowdown (i.e., a drop in $ExpShock_{ct-1}$) leads to a significant increase in the crime rate. The OLS result underestimates the impact of the export slowdown, suggesting that uncontrolled factors in the error term are positively correlated with $ExpShock_{ct-1}$, thereby dampening the magnitude of β_1. This discrepancy is likely due to omitted variable bias. For instance, an unobserved productivity shock or automation could boost exports while increasing crime rates by displacing workers, thus reducing the coefficient estimate for β_1. Similarly, a domestic demand boom might have reduced both exports and crime rates.[22] Taking the IV

[22]Bias due to reverse causality may work in the opposite direction. For example, cities with lower crime rates may attract more investment, thereby increasing exports. An OLS estimation without correcting for

;

Table 2: Baseline Results: Effects of Export Slowdowns on Crime Rate

Dep. Var.:	$\triangle$ Criminal cases per thousand$_{ct}$						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	OLS	IV	IV	IV	OLS-RF	OLS-RF	OLS-RF
ExpShock$_{ct-1}$	0.006	-0.143***	-0.193***	-0.194***			
	(0.010)	(0.051)	(0.061)	(0.067)			
ExpShock$^{IV}_{ct-1}$					-0.332***	-0.399***	-0.410***
					(0.108)	(0.111)	(0.110)
$\triangle$Log GDPpc$_{ct-1}$			-0.011	0.014		-0.051	-0.019
			(0.052)	(0.060)		(0.045)	(0.047)
$\triangle$Log Population$_{ct-1}$			0.367*	0.265		0.153	0.143
			(0.217)	(0.185)		(0.109)	(0.103)
$\triangle$Log FDI$_{ct-1}$			-0.003	0.002		-0.001	0.001
			(0.005)	(0.004)		(0.004)	(0.004)
$\triangle$College share$_{ct-1}$			-0.009	2.733**		0.679	2.194**
			(1.700)	(1.367)		(1.056)	(0.981)
Log GDPpc$_{ct-1}$			-0.050	0.069		-0.024	0.077**
			(0.036)	(0.053)		(0.026)	(0.037)
Log Population$_{ct-1}$			-0.720**	-0.550**		-0.442***	-0.373**
			(0.279)	(0.268)		(0.152)	(0.151)
Log FDI$_{ct-1}$			-0.001	-0.010*		-0.001	-0.007
			(0.006)	(0.006)		(0.005)	(0.005)
College share$_{ct-1}$			-3.112*	-4.926***		-2.117	-3.476***
			(1.732)	(1.345)		(1.400)	(0.933)
City FE	Y	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	N	Y	Y	N
Prov-year FE	N	N	N	Y	N	N	Y
First-stage F-stat	-	14.60	12.22	9.09	-	-	-
Observations	2,229	2,229	1,904	1,867	2,229	1,904	1,867
R-squared	0.489	-	-	-	0.495	0.516	0.725

Notes: Observations are at the city-year level for years from 2011 to 2018. The dependent variable is the change in the number of crimes per thousand population in city c between year t and $t-1$. The main explanatory variable $ExpShock_{ct-1}$ and the Bartik IV $ExpShock^{IV}_{ct-1}$ are constructed according to equations (1) and (3), respectively. Column (1) reports OLS estimates, while columns (2) to (4) are results from IV regressions. Columns (5) to (7) report results from reduced-form IV regressions. Additional time-varying city-level controls in columns (3) to (4) and (6) to (7) include both the changes between year $t-1$ and $t-2$ and the levels. All control variables are in logs except for the college share. We include city fixed effects and year fixed effects in columns (1) to (3) and columns (5) to (6), and replace year fixed effects with the province-year fixed effects in columns (4) and (7). Robust standard errors are reported in parentheses and are clustered at the city level. *** p<0.01, ** p<0.05, * p<0.1.

estimation in column (2) as the baseline, a reduction of \$1,000 in a city's export per worker would induce 143 additional criminal cases per million population. This effect is considerable, given that the average crime rate in our sample is 447 cases per million population. A back-of-the-envelope calculation suggests that nearly 13.1% of the interquartile difference in crime rates could be explained by the difference in their export levels.[23]

Our baseline findings remain robust when we include a series of time-varying city characteristics from the China City Statistical Yearbooks. In column (3), we control for the log of GDP per capita, foreign direct investment, population, and the share of the college-educated population, both in first differences and in levels.[24] Column (4) demonstrates that the effect of export shocks on raising crime rates remains significantly negative even after controlling for province-year fixed effects. In all three specifications of the IV regressions, the first-stage F-statistic is at or above 10. The corresponding first-stage regressions are presented in Appendix Table A.3. Finally, in columns (5)-(7), we directly regress the change in crime rates on the constructed IV, confirming that foreign import demand has a significant and negative effect on a city's crime rates.[25]

4.2 Heterogeneity Analysis

The granularity of the data allows us to examine the heterogeneous effects of the export slowdown on crime incidents. We focus on the 2SLS specification with fixed effects, as shown in column (2) of Table 2. First, we divide the sample by the median value of the young population share, defined as individuals aged between 18 and 30 in the 2010 Population Census. We run separate regressions for cities with high and low young population shares, reporting the results in columns (1) and (2) of Table 3. The negative impact of the export slowdown on crime rates is most significant in cities with a high share of the young population, while the effect on "older cities" is less significant. This finding is consistent with empirical evidence showing a positive correlation between the young population and crime rates (Freeman, 1991; Grogger, 1998; Khanna et al., 2021).

Second, cities can differ substantially in how they accommodate migrant workers, who are

this type of bias tends to overestimate the impact of the export shock on crime rates.

[23]Alternatively, according to the estimates, had China maintained its pre-crisis annual export growth rate of 20%, the crime rates would have decreased by 12.1% from the current level.

[24]For GDP per capita and the share of the college-educated population, we normalize using contemporary population data. This approach enables us to more accurately capture the city-year-level economic and demographic factors that may influence crime rates.

[25]To ensure that missing information on the occurrence time and location of criminal activity does not affect our baseline results, we regress the city-level average missing rate from 2011 to 2018—measured by the number of missing cases as a share of total crime cases—against our explanatory variable of interest, the average city-level export shock. The coefficient for the export shock is insignificant, with a magnitude of -0.031 and a standard error of 0.022.

Table 3: Heterogeneous Effects of Export Slowdowns by Initial City Characteristics

Dep. Var.:	△ Criminal cases per thousand$_{ct}$					
	Young share		Migration share		Mfg. employment share	
	High	Low	High	Low	High	Low
	(1)	(2)	(3)	(4)	(5)	(6)
ExpShock$_{ct-1}$	-0.163**	-0.074	-0.140**	0.105	-0.211**	-0.064
	(0.067)	(0.062)	(0.063)	(0.224)	(0.083)	(0.074)
Mean crime rate	0.519	0.374	0.581	0.311	0.536	0.372
Test for equal coeff.	p = 0.286		p = 0.074		p = 0.239	
City FE	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y
First-stage F-stat	8.39	6.56	9.36	0.77	7.09	6.90
Observations	1,117	1,112	1,100	1,129	978	1,003

	Dropout share		By regions			Initial crime level	
	High	Low	East	Middle	West	High	Low
	(7)	(8)	(9)	(10)	(11)	(12)	(13)
ExpShock$_{ct-1}$	-0.247***	-0.099	-0.151**	-0.022	-0.045	-0.122**	-0.124
	(0.079)	(0.075)	(0.070)	(0.059)	(0.059)	(0.058)	(0.103)
Mean crime rate	0.481	0.418	0.557	0.387	0.404	0.621	0.272
Test for equal coeff.	p = 0.192		p = 0.277, p = 0.279			p = 0.498	
City FE	Y	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y	Y
First-stage F-stat	10.27	3.93	7.35	5.00	6.58	8.79	5.70
Observations	922	902	684	714	831	1,099	1,130

Notes: Observations are at the city-year level for the year 2011 to 2018. The dependent variable is the change in the number of crimes per thousand population in city c between year t and $t-1$, while the main explanatory variable $ExpShock_{ct-1}$ and the Bartik IV $ExpShock^{IV}_{ct-1}$ are constructed according to equations (1) and (3) respectively. We run separate regressions for the cities with 50% most and least young share in columns (1) and (2), migration share in columns (3) and (4), manufacturing employment share in columns (5) and (6), dropout share in columns (7) and (8), and initial crime rate in columns (12) and (13). The regional split in columns (9) to (11) is East, Middle and West according to the division provided by the National Development and Reform Commission. Data on young share, migration share and dropout share comes from the 2010 Census. Data on manufacturing employment comes from China City Statistical Yearbooks. All columns report IV estimates and include city fixed effects and year fixed effects. Robust standard errors are reported in parentheses and are clustered at the city level. *** p<0.01, ** p<0.05, * p<0.1.

often low-skilled, receive lower wages, and have inadequate social safety nets, making them highly susceptible to negative economic shocks. Research indicates that these workers face significant economic and social marginalization, leading to increased stress and desperation (Zhong et al., 2016). This vulnerability increases their likelihood of engaging in criminal activities as a means of coping with or mitigating their precarious economic conditions. In columns (3) and (4) of Table 3, cities are categorized according to their initial migration share. Unsurprisingly, we observe a negative and significant impact of the export shock in cities with a higher migration share. For cities with a lower-than-median migration share, the effect remains negative but is no longer significant. The finding is in line with the observation that the migrant population tends to be younger than the local population and supports previous studies showing that low-skilled international migrant workers are more likely to commit crimes (Rattner, 1997; Aoki and Todo, 2009; Baker, 2015; Caminha et al., 2017).

Third, we divide cities by the median level of manufacturing employment share in 2010.[26] As most exports from China are manufactured goods, we expect that the export slowdown more severely affects cities with higher manufacturing share. Results reported in columns (5) and (6) confirm this point: the export slowdown has a stronger impact on cities with a higher initial share of manufacturing employment.

In columns (7) and (8), we investigate whether cities with a higher dropout share witness more occurrences of criminal activities. The dropout share is computed as the proportion of children aged 7-18 who are not in school, using data from the 2010 Population Census. The estimated results show more pronounced effects of the export shocks in cities with a higher dropout share, consistent with the findings of Dix-Carneiro and Kovak (2017).

The impact of the export slowdown on crime rates may also differ geographically. Thus, in columns (9)-(11), we divide cities by their geographical location. We categorize cities into three regions: east, middle, and west.[27] The results show that export shocks make significant changes in crime rates in the east regions but not in the middle and the west. This pattern is consistent with the fact that manufacturing production in China was highly concentrated in eastern coastal provinces.

In the last two columns, we divide the sample at the median level of crime rate in 2014 to explore whether the export slowdown has a bigger impact in cities where the initial crime

[26] The manufacturing employment share is calculated as the proportion of manufacturing employment in total employment, using the 2010 China City Statistical Yearbooks.

[27] The east region includes provinces of Beijing, Tianjin, Hebei, Liaoning, Shanghai, Jiangsu, Zhejiang, Fujian, Shandong, Guangdong, and Hainan;the middle region includes provinces of Shanxi, Jilin, Heilongjiang, Anhui, Jiangxi, Henan, Hubei, and Hunan; the west region includes provinces of Sichuan, Chongqing, Guizhou, Yunnan, Xizang, Shaanxi, Gansu, Qinghai, Ningxia, Xinjiang, Guangxi, and Inner Mongolia.

Table 4: Heterogeneous Effects of Export Slowdowns by Crime Type

Dep. Var.:	△ Criminal cases per thousand$_{ct}$						
	(1) IV	(2) IV	(3) IV	(4) IV	(5) IV	(6) IV	(7) IV
Type of crime	Public security	Felony traffic offense	Violence	Robbery	Stealing	Defraud	Extortion
ExpShock$_{ct-1}$	-0.002	-0.032*	-0.020**	-0.007*	-0.025*	-0.006	-0.001
	(0.001)	(0.017)	(0.009)	(0.004)	(0.014)	(0.005)	(0.001)
Mean crime rate	0.008	0.107	0.068	0.008	0.082	0.019	0.002
City FE	Y	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y	Y
First-stage F-stat	14.60	14.60	14.60	14.60	14.60	14.60	14.60
Observations	2,229	2,229	2,229	2,229	2,229	2,229	2,229
	(8) IV	(9) IV	(10) IV	(11) IV	(12) IV	(13) IV	(14) IV
Type of crime	Drugs	Prostitution	Gambling	Counterfeiting	IP	Finance	Bribery
ExpShock$_{ct-1}$	-0.007	-0.005*	-0.006*	0.002	-0.0001	0.0003	-0.001
	(0.015)	(0.003)	(0.003)	(0.003)	(0.001)	(0.001)	(0.001)
Mean crime rate	0.042	0.003	0.009	0.003	0.001	0.002	0.004
City FE	Y	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y	Y
First-stage F-stat	14.60	14.60	14.60	14.60	14.60	14.60	14.60
Observations	2,229	2,229	2,229	2,229	2,229	2,229	2,229

Notes: Observations are at the city-year level for years from 2011 to 2018. The dependent variable is the change in the number of crimes per thousand population in city c between year t and $t-1$. The classification of crime is described in Appendix Table A.1. The main explanatory variable $ExpShock_{ct-1}$ and the Bartik IV $ExpShock^{IV}_{ct-1}$ are constructed according to equations (1) and (3), respectively. All columns report IV estimates and include city fixed effects and year fixed effects. Robust standard errors are reported in parentheses and are clustered at the city level. *** p<0.01, ** p<0.05, * p<0.1.

rates are relatively high. Results confirm this point.[28]

Notably, a significant coefficient estimate only indicates that the effect is present and significantly different from zero within its respective group; it does not provide information about whether the effects differ across groups. By focusing on cities with higher proportions of young people, migrants, manufacturing activities, school dropouts, initial crime rates, or those located in the eastern region, our objective is to understand the underlying factors contributing to the increase in crime rates.[29]

Table 4 examines how the impact of the export slowdown on crime rates varies by type of crime. As explained in section 2.1, we categorize criminal cases into fourteen groups based on the Criminal Law of the People's Republic of China. This classification is derived from a textual analysis of keywords included in the case titles and contents.[30] The results show that

[28]We use data from 2014 instead of 2010 to calculate the initial crime rates and split the sample, as the CJO was established at the end of 2013, making the pre-2014 sample less complete.

[29]In Table 3, we follow Cleary (1999) and use a bootstrapping method to calculate the empirical p-value to test whether the coefficients from different subgroups are significantly different from each other. This method estimates the likelihood of obtaining the observed differences in coefficient estimates between groups if the true coefficients are indeed equal. We find that the differences in the coefficient estimates are insignificant, except for the two groups divided by the initial share of migrants.

[30]The full list of keywords in Chinese, along with their English translations, is provided in Appendix Table

a negative export shock raises both economically motivated and impulsive criminal activities, such as felony traffic offenses, violence, robbery, stealing, prostitution, and gambling.

Overall, the export slowdown disproportionately affects crime rates in cities specializing in manufacturing, with larger populations of young people, migrants, dropouts, or higher initial crime rates. Individuals in these cities are more prone to engaging in criminal activities such as felony traffic offenses, violence, prostitution, gambling, and resource appropriation crimes like robbery and stealing. This evidence supports the broader narrative that weakening labor market conditions from declining exports lower the opportunity costs of committing crimes for economically vulnerable individuals. Additionally, the pronounced effects on non-economically motivated crimes may be driven by psychological factors, such as stress caused by worsening economic conditions, which we will explore in section 5.

4.3 Robustness Checks

We perform a battery of robustness checks for our baseline results. In the following tables, we use the same specification as in column (2) of Table 2.

4.3.1 Basic Robustness Checks

Measurement issue in crime rates using CJO data: We conduct several checks to ensure the representativeness and accuracy of crime data from the CJO and provide evidence that measurement issues in CJO-based crime rates do not affect our results. First, we use alternative samples to verify that our estimation results are not influenced by the less complete case reporting in the early years. In columns (1) and (2) of Appendix Table A.5, we replicate the baseline regression using criminal cases committed or filed after 2014.

Furthermore, to ensure that changes in the crime rate are not driven by changes in the conviction rate due to varying economic opportunities, we analyzed the number and share of criminal cases resulting in acquittals. In the Chinese context, the conviction rate is remarkably high. Our statistics show that the share of acquittals is only 1.19% during our sample period, with a slight increase from 0.82% to 1.09% between 2010 and 2018. Columns (3) and (4) of Appendix Table A.5 present the regressions estimating the impact of the export slowdown on acquittal rates. Column (3) focuses on the changes in the number of criminal cases with acquittals per million population, while column (4) examines the changes in acquittals as a share of the total number of sentenced cases each year. The findings indicate that adverse export shocks do not significantly affect acquittal rates.

Another aspect of the Chinese court system is that cases, particularly minor ones, can be resolved through mediation. According to related provisions in Chinese laws, mediation

A.1. Appendix Table A.4 presents the number and share of each type of criminal case. Felony traffic offenses account for the largest proportion, followed by cases related to theft, violence, drugs, and defrauding.

is generally restricted to cases deemed less harmful to society or with lower social costs.[31] A detailed analysis of our data shows that the share of cases resolved through complete out-of-court settlements is minimal, at only 0.02%, while cases involving partial mediation, where the victim and the accused person reach an agreement on the civil aspects, account for about 5.9%. To determine if the export slowdown affects the likelihood of mediation, we regress the city-specific mediation rate on export shocks, with the mediation rate defined as the share of mediation cases. Columns (5) and (6) of Appendix Table A.5 indicate that the export slowdown does not significantly impact the probability of mediation across cities.

Last, our crime data only includes cases where an offender has been identified and taken to court, meaning enforcement levels could affect reported crime rates. Lacking data on "reported crime," we follow economic literature using internet search intensity (e.g., Google Trends) to gauge public interest in socioeconomic issues (Madestam et al., 2013; Stephens-Davidowitz, 2014; Kearney and Levine, 2015). We use the Baidu Index to search for keywords related to criminal activities.[32] The search intensity of crime-related activities reflects households' perceptions of local security. The Baidu Index distinguishes searches by their city of origin. Using this measure of security concern, we regress the log change in the Baidu search index on export shocks (see Appendix Table A.7). Column (1) uses the sum of the Baidu Index for five general crime-related keywords, while column (2) uses the sum for all keywords.[33] The results show that cities more exposed to export slowdowns see a greater increase in search indices for crime-related keywords. Columns (3) to (14) further analyze specific crime categories, revealing that negative export shocks lead to higher search intensity for crimes related to violence, robbery, drugs, prostitution, gambling, and bribery, consistent with our baseline findings in Table 4.

Influential observations: Next, in column (1) of Appendix Table A.8, we replace the year-fixed effects with region-year fixed effects. To ensure our results are not driven by

[31] According to the China Criminal Procedure Law, mediation is permissible in specific situations, reflecting the law's approach to resolving less harmful or lower-cost social cases. Mediation is allowed when the People's Court hears cases of incidental civil litigation (i.e., Article 103) and in private prosecution cases where the victim can sue directly in court, particularly for minor criminal offenses (i.e., Article 212). Mediation can also be applied in public prosecution cases arising from civil disputes involving offenses under Chapter IV (i.e., Crimes of Infringing upon Citizens' Right of the Person and Democratic Rights) and Chapter V (i.e., Crimes of Property Violation) of the Criminal Law, provided these offenses may result in sentences of less than three years of imprisonment (i.e., Article 288). Finally, mediation is also allowed in other public prosecution cases, excluding malpractice offenses, if the potential punishment is less than seven years' imprisonment (i.e., Article 288).

[32] Appendix Table A.6 provides the full list of keywords in Chinese and their English translations. Baidu, the largest search engine in China, accounts for about 60-70% of internet searches.

[33] Previous research has confirmed that the Baidu Index is linearly correlated with the volume of public searches for a given keyword (Qin and Zhu, 2018). By summing the Baidu Index values of different keywords, we can capture the overall search intensity related to a specific topic.

the winsorization of $ExpShock_{ct}^{IV}$, we alternatively winsorize it at the 0.5th and 99.5th percentiles in column (2) and at the 0.1th and 99.9th percentiles in column (3). Both estimations remain negative and statistically significant. To test the influence of very large provinces, we replicate the baseline estimation 31 times, each time excluding one province. We report the maximum and minimum coefficient estimates in column (4). In all cases, the estimates are consistently negative and significant. To visualize these results, we present in Appendix Figure A.3 the residual scatter plots for both the first-stage and the second-stage regressions in column (2) of Table 2. In the right panel, the horizontal axis presents residual terms obtained from regressing the predicted export shocks in the first stage against variables on the right-hand side of equation (2) excluding $ExpShock_{ct}$. The vertical axis presents residual terms from regressing the change in crime rates against the right-hand side variables in equation (2), again excluding $ExpShock_{ct}$. In the left panel, the vertical variable is the residualized $ExpShock_{ct}$ and the horizontal variable is the residualized $ExpShock_{ct}^{IV}$. The left panel shows a strong positive correlation between the two variables, consistent with the high F-statistic for the first-stage regression. The right panel reveals a downward-sloping relationship and shows no significant outliers. Moreover, Appendix Figure A.4 presents the residual binned scatter plots, where city-year level observations are grouped into 50 bins based on their residualized export shocks. Both panels show a close relationship between the vertical and horizontal variables. Notably, the binned regression estimates a coefficient of -0.11 with a t-value at -2.24, after removing the outlier bin on the left tail.

4.3.2 Other City-level Shocks and Incidents

One concern for our identification is that the demand shocks that originated from the OECD countries can be correlated with internal shocks within Chinese cities. To address this concern, we construct several measures to capture the internal shocks and include them as additional controls in equation (2). To be specific, we consider three possible internal shocks that are related to domestic demand, domestic supply, and import demand, respectively. First, we use the change in domestic absorption (i.e., domestic output less net exports, $Absorption_{jt} = Output_{jt} - Export_{jt} + Import_{jt}$) to proxy for domestic demand shock.[34] We apply the same shift-share approach, using each city's initial industrial specialization as weights, to measure the demand shock to each city:

$$AbsorptionShock_{ct} = \Sigma_j \frac{L_{cj,2010}}{\Sigma_c L_{cj,2010}} \frac{\Delta Absorption_{jt}}{L_{c,2010}}. \tag{4}$$

[34] For industry, we use the four-digit Chinese Standard Industrial Classification (CSIC). Data on industry output comes from China Industry Statistical Yearbooks. Initial specialization structure is calculated using industrial employment, base on the 2010 China Annual Survey of Industrial Firms. Data on exports and imports is from China Customs Database. We use a concordance to match the HS 6-digit codes with the CSIC 4-digit industries.

Likewise, we use the change in industrial output to measure the domestic supply shock, where each city's exposure to the supply shock also depends on the initial employment structure:

$$OutputShock_{ct} = \Sigma_j \frac{L_{cj,2010}}{\Sigma_c L_{cj,2010}} \frac{\Delta Output_{jt}}{L_{c,2010}}. \tag{5}$$

To assess the potential confounding effect of import competition, we measure each city's exposure to the import shock using the same shift-share formula:

$$ImpShock_{ct} = \Sigma_j \frac{L_{cj,2010}}{\Sigma_c L_{cj,2010}} \frac{\Delta M_{jt}}{L_{c,2010}}. \tag{6}$$

Columns (1) to (3) of Appendix Table A.9 present the results as we gradually control for domestic demand shocks, output shocks, and import shocks. In all cases, the estimated export shock coefficient remains negative and statistically significant.

Spatial correlation across cities in the export shocks can also confound the interpretation of our results. To address this concern, we construct a weighted average of the export shocks across all cities that share an administrative border with city c, using their working-age population as weights. This measure controls for the impacts of export shocks experienced by the neighboring cities. We include it as an additional control in equation (2), and use the weighted average of $ExpShock^{IV}$ as the instrument. The result is reported in column (4) of Appendix Table A.9. Only the local export shocks have a negative effect on the crime rates.

To further ensure that the observed changes in crime rates due to export slowdowns are not driven by incidents such as labor strikes as discussed in Campante et al. (2023), we incorporated data from their study and included the change in the number of labor strike events per thousand population as a control variable in our regression analysis. Column (5) of Appendix Table A.9 displays the result. Although labor strikes have marginally pronounced effect on local crime rates, the coefficient for $ExpShock_{ct-1}$ remains significant at the 1% level.

To rule out the possibility that unobserved stochastic trends drive our findings, we randomly assign $ExpShock$ to cities and repeat the IV regressions 500 times. Appendix Figure A.5 shows the distribution of estimates from these simulations, with our baseline estimate marked by a red vertical line. The simulated estimates center around zero, while our benchmark estimate is beyond the 95th percentile.

4.3.3 Validity of the Shift-Share Instrument

The validity of our identification strategy hinges on the instrument reflecting aggregate foreign demand in export destinations while being independent of Chinese firms' export decisions. Given its shift-share structure, this requires that, conditional on year and city-fixed effects, other unobserved factors in the error term are uncorrelated with (1) product-specific foreign demand shocks at the national level (i.e., the shifts) and (2) initial city-product-specific export shares (i.e., the shares). We conduct several checks to test these assumptions and show that our estimates are free from pre-trend issues and pass other statistical tests recommended by the shift-share literature.

Exogeneity of the shifts: We first test for assumption (1). The exogeneity of the shifts can be violated if some products with stronger demand shocks tend to be concentrated in cities with certain characteristics that are also related to the crime rate. To address this issue, we run a balance test as suggested by Borusyak et al. (2022). In particular, we examine whether our product-level shifts are correlated with the exposure-weighted average of initial city-level characteristics.

Our shift-share IV can be written in the general form of $\Sigma_k s_{ck} g_{kt}$. In our setting $g_{kt} = \frac{\Delta X_{kt}}{\Sigma_c X_{ck,2010}}$ is the demand shock of product k from the foreign market divided by the total export of product k from China in 2010. And $s_{ck} = \frac{X_{ck,2010}}{L_{c,2010}}$ is city c's export of product k divided by its initial working-age population, which measures city c's exposure to the shock to product k. Borusyak et al. (2022) prove that the validity of the instrument relies on:

$$\Sigma_k s_k g_{kt} \phi_k \xrightarrow{p} 0, \tag{7}$$

where $s_k = E(s_{ck})$ is the expectation of the exposure to product k's shock across cities, and $\phi_k = \frac{E(s_{ck}\epsilon_c)}{E(s_{ck})}$ is the expectation of a certain initial characteristic (ϵ_c) of city c, weighted by its exposure measure s_{ck}. Equation (7) suggests that when weighted by s_k, the correlation between g_{kt} and ϕ_k approaches zero in a large sample. This implies that product-level demand shocks from the foreign market are effectively randomly assigned across cities. To verify this property, we regress g_{kt} on ϕ_k and year dummies using weighted least squares with s_k as weights. Besides the initial city-level crime rate $(Crime/L)_{c,2010}$, we also include in ϕ_k a set of predetermined city characteristics: the share of the population with a college degree, manufacturing employment share, exports as a share of GDP, the share of the population without Hukou registration, GDP per capita, fiscal expenditure per capita, fiscal revenue per capita, the share of the young population, and shocks related to domestic absorption, output, and imports, constructed according to equations (4), (5), and (6), respectively. Appendix Table A.10 reports the coefficients and standard errors for each city-level variable, as well as the joint χ^2 statistic. The coefficients are all statistically insignificant, indicating that our product-level export shocks meet the requirement of treatment balance.[35]

[35] In our sample, the Herfindahl index based on the initial export structure in 2010 is 0.0056, implying an effective sample size of 1420 (i.e., 8/0.0056). To compute the Herfindahl index, we re-normalized the expectation of the exposure measure s_k so that they sum to one. Given that our panel covers 8 years, the effective sample size is determined by 8/HHI.

Exogeneity of the shares: We next examine the assumption (2): the initial city-product-specific export share (i.e., the share) is not correlated with the unobserved error. As highlighted by Goldsmith-Pinkham et al. (2020), the assumption that the Bartik shares are exogenous can be violated if the city's exposure to export shocks is driven by a small number of products that experience particularly large shocks or predetermined trends. We implement the check suggested by Goldsmith-Pinkham et al. (2020) and compute the Rotemberg weights for export shocks with city- and year-fixed effects. The product-specific Rotemberg weights capture the degree of sensitivity to mis-specifications when the exogeneity assumption of initial export shares fails. The 2SLS estimates are more sensitive to the bias introduced by a product with a large Rotemberg weight. In Appendix Table A.11, we report five 6-digit HS products with the highest Rotemberg weights. The top five products account for 8.2% of the positive weights in the shift-share estimator, which is far lower than the cases studied in Goldsmith-Pinkham et al. (2020).

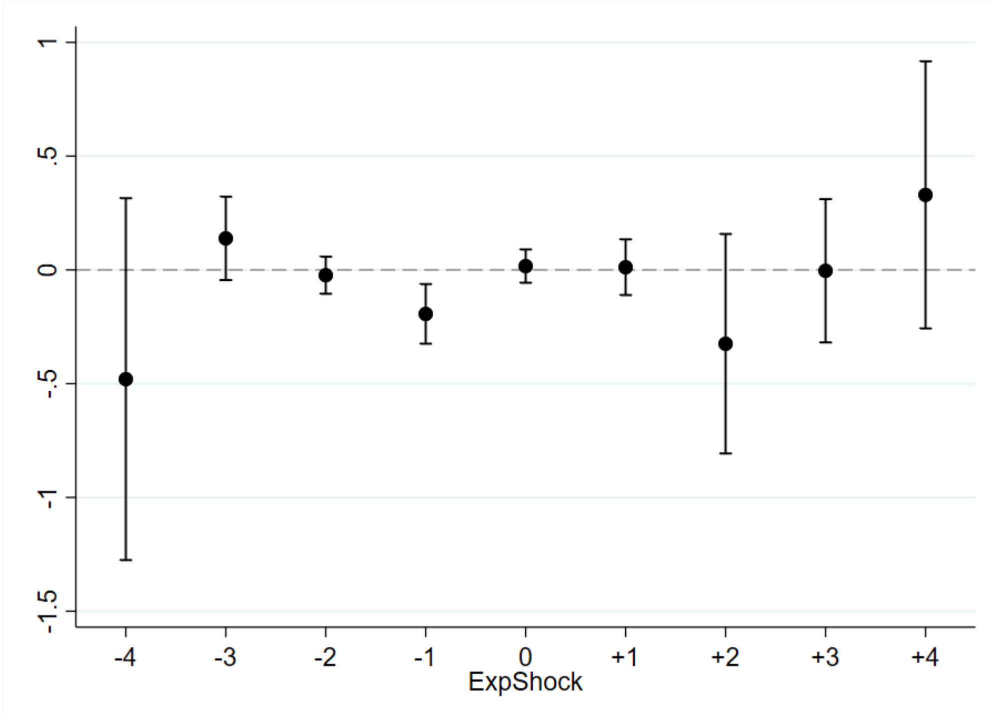
We then adopt the method of dropping one HS section at a time to assess whether our estimation relies on large shocks to any particular products. We re-run the 2SLS regressions excluding one HS section at a time, and the HS sections and their corresponding HS codes are listed in Table A.12 in the Appendix. We report the maximum and minimum coefficient estimates with their standard errors in column (1) of Appendix Table A.13. Both the maximum and the minimum estimates remain highly significant and negative, suggesting that our baseline results are not driven by particular products.

4.3.4 Pre-trend Tests

First, we regress the change in city crime rates in year t on different lags and leads of export shock measures, one at a time. Figure 4 reports the point estimates with 95% confidence intervals, where "-1" represents the one-year lag of export shocks, "0" denotes current export shocks, and "+1" indicates the one-year lead of export shocks, with similar representations for other labels. The figure reveals that the coefficient for the export shock measure lagged by one year significantly affects crime rates, while leading export shocks do not influence changes in crime rates in the current period, as indicated by the non-significant estimates for the coefficients of all leads of export shock measures.

As a second pre-trend test, we collected provincial-level crime data from the China Procuratorial Yearbook and measured the crime rate as the number of arrests per thousand population for each year from 2001 to 2009, following Edlund et al. (2013). Using these data, we regressed the change in crime rate, as reported in the CPY from 2002 to 2009, on the nine-year lead of export shocks from 2011 to 2018. The results are presented in column (2) of Appendix Table A.13. The coefficient for export shock measures is statistically insignifi-

Figure 4: Impact of Lags and Leads of Export Shocks on Crime Rates



Notes: The figure presents the point estimates with 95 percent confidence intervals for each lag and lead of export shocks. In each regression, we control for covariates both in first-differences and levels, and include provincial-year and city fixed effects.

cant, indicating no impact of future export slowdowns on past crime rates.[36] These findings suggest that pre-existing trends in city crime rates are unlikely to bias our main results.

4.3.5 Other Statistical Tests

Alternative comparator countries with export structures resembling China: In our baseline regression, we proxy foreign demand shocks using the change in imports of twenty-eight OECD countries from the top nine countries with export structures most similar to China's. To ensure our results are robust and not driven by the specific selection of these comparator countries, we also use the by-product imports of OECD countries from the top five and top fifteen nations whose export structures closely resemble China's to capture the change in foreign demand shocks. Results are shown in columns (3) and (4) of Appendix Table A.13, and the coefficient of export shock remains negative and statistically significant.

Alternative source of variations: To utilize more detailed information of the destination market, we construct an alternative measure of $ExpShock_{ct}^{IV}$ based on the destination-specific

[36]We also tests for the pre-trend using leads of $t+1$, $t+2$, $t+3$ and $t+4$ with provincial-level data. All coefficients are statistically insignificant, further confirming the absence of pre-trends.

demand shocks:

$$ExpShock_{ct}^{IV} = \Sigma_k \Sigma_d \frac{X_{cdk,2010}}{\Sigma_c X_{cdk,2010}} \frac{\Delta X_{dkt}}{L_{c,2010}}, \tag{8}$$

where X_{dkt} denotes country d's imports of product k from top nine countries in year t. The share term $\frac{X_{cdk,2010}}{\Sigma_c X_{cdk,2010}}$ reflects the importance of city c in China's total exports of product k to destination d in 2010. Compared with equation (3), the alternative measure of foreign demand shock combines more disaggregate information on the destination market and each city's relative market share in each destination market. The results are reported in column (5) of Appendix Table A.13, showing very close estimate to the baseline regression.

Alternative statistical inference: Finally, we evaluate the robustness of our results by clustering the standard errors at several alternative levels. In column (1) of Appendix Table A.14, we cluster the standard errors at the province level to allow for the existence of within-province across-city correlation. As pointed out by Adao et al. (2019), cities located in different provinces may have similar industry/export structure and therefore experience correlated shocks. To take this possible correlation into consideration, we follow Campante et al. (2023) and compute the similarity of city c's initial export structure with that of the provincial capital city j based on the index proposed by Finger and Kreinin (1979):

$$SimilarityIndex_{cj} = \Sigma_k min \left\{ \frac{X_{ck}}{X_c}, \frac{X_{jk}}{X_j} \right\}, \tag{9}$$

where X_{ck} (X_{jk}) denotes the export of product k originating from city c (j) to the ROW in year 2010, and X_c (X_j) is the total exports of city c (j). This index takes values between 0 and 1, and a higher value indicates a higher degree of similarity between two cities in their export structure. In practice, we assign each city to an export-similarity cluster corresponding to the provincial capital with highest $SimilarityIndex$.[37] We eventually create 31 clusters, and each cluster has 11 cities on average. Column (2) of Appendix Table A.14 reports the results with errors clustered at the export-similarity group. In column (3), we re-assign each city to an export-similarity cluster corresponding to the provincial capital outside of its own province with which its export profile shows the highest similarity. In columns (4) and (5), robust standard errors are two-way clustered at the province and the export-similarity group level, and at the province and the outside-of-province export-similarity group level, respectively. All baseline results remain robust to these alternative clustering specifications.

In columns (6) to (9) of Appendix Table A.14, we adopt the approach by Conley (1999) to calculate the Heteroscedasticity and Autocorrelation Consistent (HAC) standard errors,

[37] We can not directly implement Adao et al. (2019)'s procedure to correct the standard errors since there are more product-level trade shocks (>4,000 HS6-digit code) than geographic units (325 cities) in our sample.

accounting for spatial correlation in the errors across neighboring regions within specified distance thresholds. We experimented with various thresholds, specifically 50, 100, 150, and 200 kilometers. The results show that the coefficient estimates for export shocks remain significant, suggesting that our basic findings are robust to the inclusion of spatially correlated standard errors.

5 Mechanism Analysis

We explore several explanations for why the export slowdown has led to a rise in crime across Chinese cities. We examine whether the increase in crime rates is linked to (1) worsening labor market conditions, (2) deteriorating mental health among the workforce, and/or (3) decreased local government expenditure on stability maintenance. Our analysis uses the specification from column (2) of Table 2. Panel B of Appendix Table A.2 provides the summary statistics for additional variables used in the mechanism analysis.

5.1 Employment and Earning

Declining employment and income opportunities lower the opportunity cost of committing crimes, thereby increasing criminal activities. First, we estimate how export shocks affect employment at the city level. The results in column (1) of Table 5 indicate that cities experiencing more severe export slowdowns see a greater reduction in total employment. Columns (2) and (3) further show that both the manufacturing and service sectors witness employment declines, with the effects being more pronounced in the manufacturing sector.

Given the large population of migrant workers in China, we next consider how deteriorating labor market conditions affect migration patterns. We compute migration flows for each city using micro-data from the 2010 and 2015 Chinese Population Census.[38] Columns (4) to (7) report the results. Columns (4) and (6) use the log change of city-level migration inflows and outflows, while columns (5) and (7) use the change in the share of migration inflows and outflows relative to the initial population in 2010. Cities more exposed to negative manufacturing export shocks tend to have a lower increase in migration inflows (although insignificant) and a higher rise in migrant outflows.

Finally, in columns (8) and (9), we present evidence that the export slowdown decreases job availability. Specifically, we construct job flow measures following Davis and Haltiwanger (1992) and Ma et al. (2015) using firm level employment information from the National

[38]Following Imbert et al. (2022), we only include migrants who were 15 to 64 years old at the time of migration. For details on the Chinese Population Census and the measurement of internal migration, see Data Appendix C.

Table 5: Mechanism: Export Shocks and Employment

Dep. Var.:	△ Labor market condition measures$_{ct}$								
	(1) Log Empl.	(2) Log Mfg. empl.	(3) Log Svc. empl.	(4) Log Migrant-in	(5) Migrant-in / 100 pop	(6) Log Migrant-out	(7) Migrant-out / 100 pop	(8) Job creation	(9) Job destruction
ExpShock$_{ct}$	0.120***	0.182***	0.077**	0.023	1.337	-0.346*	-0.345*	0.025	-0.052*
	(0.046)	(0.067)	(0.037)	(0.217)	(0.973)	(0.184)	(0.202)	(0.024)	(0.027)
City FE	Y	Y	Y	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y	Y	Y	Y
First-stage F-stat	17.49	17.49	17.49	13.35	13.57	13.84	13.53	5.87	6.71
Observations	2,265	2,265	2,265	1,497	1,592	1,570	1,587	1,171	1,206

Notes: Observations are at the city-year level. Dependent variables in columns (1) to (3) are the log change in total employment, manufacturing employment and service employment in year t respectively, which are computed from the China City Statistical Yearbooks (2011 - 2018). Dependent variables in columns (4) and (5) are the log change in total migration inflows and migration inflows per hundred workers in year t. Dependent variables in columns (6) and (7) are the log change in total migration outflows and migration outflows per hundred population in year t. Migration flows are computed from the 2010 and 2015 1% Population Census. Dependent variables in columns (8) and (9) are the job creation rate and job destruction rate constructed following Davis and Haltiwanger (1992), which are computed from the National Tax Survey (2010 - 2015). The main explanatory variable $ExpShock_{ct}$ and the Bartik IV $ExpShock_{ct}^{IV}$ are constructed according to equations (1) and (3), respectively. All columns report IV estimates, which include city- and year-fixed effects. Robust standard errors are reported in parentheses and are clustered at the city level. *** p<0.01, ** p<0.05, * p<0.1.

Tax Survey (2010-2015).[39] Results indicate that the export slowdown reduces job creation (although insignificant) while increasing job destruction.

The impact of export slowdown on labor earnings is reported in Table 6. Column (1) uses a city's overall average wage, computed from the National Tax Survey. We find that the export slowdown exerts downward pressure on overall wages, although the estimates lack statistical precision. In column (2), we investigate the wage effects on migrants using individual-level income information from the China Migrants Dynamic Survey (2011-2018).[40] We measure city-level wages of migrants by averaging the wages of all migrants residing in the city. Results in column (2) show that negative export shocks lower the labor earnings of migrants, consistent with previous findings that the migrant population is more vulnerable to negative shocks. These results align with the estimates in Table 3, which show that cities with higher migrant shares experience more pronounced increases in crime rates. In columns (3) to (6), we explore the heterogeneous effects on the labor earnings of migrants with different characteristics. We divide migrants into "Low Education" and "High Education" groups based on whether their education level was at junior secondary school or below, and compute the average wage of each group at the city level. We find that export slowdowns lower the labor income of low-educated migrants and have an insignificant effect on the higher-educated group. Finally, we distinguish employment by sector. Results are reported in

[39]The National Tax Survey is collected by China's Ministry of Finance and State Administration of Taxation. Details on measuring job creation and job destruction are provided in Appendix E.

[40]The China Migrants Dynamic Survey is collected by the National Health Commission. Details on this database are introduced in Data Appendix C.

Table 6: Mechanism: Export Shocks and Labor Income

Dep. Var.:	△ Labor market condition measures$_{ct}$					
	(1) Log Avg. wage	(2) Log Migrant wage	(3) Low Educ.	(4) High Educ.	(5) Manufacture	(6) Service
ExpShock$_{ct}$	0.263	0.088*	0.104**	0.052	0.273*	0.104**
	(0.274)	(0.046)	(0.051)	(0.051)	(0.159)	(0.050)
City FE	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y
First-stage F-stat	12.32	11.79	11.77	11.77	11.57	11.55
Observations	1,580	1,913	1,915	1,915	1,860	1.877

Notes: Observations are at the city-year level. Dependent variables in column (1) is the log change in average wage in year t, which are computed from the National Tax Survey. Dependent variables in column (2) is the log change in average wage of migrants in year t, which are sourced from the China Migrants Dynamic Survey (2011-2018). In columns (3) to (6), we further investigate the wage effects of migrants with different education background and industries. Manufacturing and service wage are winsorized at the 1st and 99th percentiles. The main explanatory variable $ExpShock_{ct}$ and the Bartik IV $ExpShock_{ct}^{IV}$ are constructed according to equations (1) and (3), respectively. All columns report IV estimates, which include city- and year-fixed effects. Robust standard errors are reported in parentheses and are clustered at the city level. *** p<0.01, ** p<0.05, * p<0.1.

columns (5) and (6) for the manufacturing and service sectors, respectively. Negative export shocks decrease wages in both sectors, but the effects are more substantial in manufacturing.

Taken together, this evidence supports the hypothesis that the export slowdown reduces employment opportunities and lowers wages, especially in the manufacturing sector. Our estimation shows that a \$1,000 reduction in exports per worker leads to a 12% decline in total employment and a 8.8% drop in the average wage of migrant workers. The export slowdown disproportionately affects migrant workers, who are often low-skilled and receive lower wages, making them highly susceptible to negative economic shocks. Previous research indicates that these workers face significant economic and social marginalization, leading to increased stress and desperation (Zhong et al., 2016). The slowdown in exports further exacerbates their situation by reducing the opportunity cost of committing crimes and thereby increasing their likelihood of engaging in criminal activities.

5.2 Individual Mental Health

Another possible explanation for the increased crime rates is the negative impact of export slowdowns on individual mental health. Increased stress and desperation from poor mental health may drive individuals to resort to criminal activities as a coping mechanism or out of frustration. Economic hardships, loss of employment, and reduced income can exacerbate these feelings, leading to higher crime rates (Colantone et al., 2019; Autor et al., 2019; Britto et al., 2022).

To explore the effects of the export slowdown on mental health, we calculate individual depression scores using data from the China Family Panel Studies (CFPS, 2010-2018). The CFPS uses the *Kessler Psychological Distress Scale* (K6) developed by Kessler et al. (2002),

Table 7: Mechanism: Export Shocks and Mental Health

Dep. Var.:	△ Mental health measures$_{ict}$				
	(1)	(2)	(3)	(4)	(5)
	K6 scores	K6 Manu.	K6 Svc.	K6 Young	K6 Old
ExpShock$_{ct}$	-0.086	-0.229**	-0.112	-0.237*	-0.077
	(0.120)	(0.103)	(0.105)	(0.141)	(0.124)
City FE	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y
First-stage F-stat	12.23	32.11	14.81	11.60	12.24
Observations	52,947	4,753	9,772	4,037	48,907

Notes: Observations are at the city-year level. The dependent variable in column (1) is the change in individuals' scores on the Kessler Psychological Distress Scale for year t, computed from the China Family Panel Studies (2010-2018). In columns (2) to (5), we further investigate changes in mental health conditions for sub-groups defined by different employment industries and ages. The main explanatory variable, $ExpShock_{ct}$, and the Bartik IV, $ExpShock_{ct}^{IV}$, are constructed according to equations (1) and (3), respectively. All columns report IV estimates, which include city- and year-fixed effects. Robust standard errors are reported in parentheses and are clustered at the city level. Statistical significance is indicated by *** p<0.01, ** p<0.05, and * p<0.1.

which is widely used in mental health research.[41] The K6 scale is a reliable measure that assesses the frequency of various negative mental states experienced over the past month. It consists of six questions: (i) How often do you feel emotionally frustrated, depressed, and unmotivated? (ii) How often do you feel nervous? (iii) How often do you feel restless and have trouble staying calm? (iv) How often do you feel hopeless about the future? (v) How often do you find it difficult to do anything? (vi) How often do you think life is meaningless? Each question has five response options scored from 0 to 4: "Never," "Sometimes," "Half of the time," "Usually," and "Almost every day." The higher the frequency of these feelings, the higher the score assigned to the response. We aggregate the scores from each question to obtain a total score that measures the overall degree of depression for each individual.

Table 7 presents the results of regressing the change in individual-level mental health measures on export shocks. Column (1) examines the impact of the export slowdown on psychological stress using the entire sample. Although a negative export shock appears to increase overall psychological stress, the coefficient remains statistically insignificant, possibly due to heterogeneity across individuals. To explore these heterogeneous effects, we divide the sample into different subgroups in columns (2) to (5). We find that the export slowdown significantly worsens mental health conditions for individuals employed in the manufacturing sector and those aged between 16 and 30. This finding is consistent with the result that cities with a higher proportion of manufacturing employment and younger populations experience greater increases in crime rates (i.e., see Table 3). These results highlight the broader social

[41] The China Family Panel Studies is collected by the Institute of Social Science Survey of Peking University. Details on this database are introduced in Data Appendix C. The K6 method has been extensively utilized in the literature to measure mental health and conduct economic studies, as evidenced by works such as Kessler et al. (2003), Kling et al. (2007), Twenge et al. (2019) and Zhang et al. (2020).

Table 8: Stability Maintenance Expenditure

Dep. Var.:	△ Log fiscal measures per capita$_{ct}$				
	(1) Total Fiscal Revenue	(2) Total Fiscal Expenditure	(3) Public Security	(4) Social Spending	(5) Stability Spending
ExpShock$_{ct-1}$	-0.040*	-0.037*	-0.044	-0.034*	-0.038*
	(0.024)	(0.019)	(0.034)	(0.020)	(0.020)
City FE	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y
First-stage F-stat	12.71	12.76	11.27	12	11.50
Observations	1,940	1,943	1,687	1,685	1,652

Notes: Observations are at the city-year level. Fiscal data comes from Fiscal Statistical Yearbooks, Province Statistical Yearbooks, and City Statistical Yearbooks. The dependent variables are the log change in fiscal measures per capita between year t and $t-1$. The fiscal measures we consider include total fiscal revenue (column 1), total fiscal expenditure (column 2), fiscal expenditure on public security (column 3), fiscal expenditure on "social spending" (column 4), and fiscal expenditure on "stability spending" (column 5), where "stability spending" is the sum of public security expenditure (which includes all expenses by the People's Armed Police, prosecutorial system, the court system, judicial system, and public security organs) and "social spending" (which is the sum of expenditure on public services, education, social security, medical services and urban and rural affairs). All dependent variables are winsorized at the 1st and 99th percentiles. All columns report IV estimates, which include city- and year-fixed effects. Robust standard errors are reported in parentheses and are clustered at the city level. *** p<0.01, ** p<0.05, * p<0.1.

and psychological impacts of economic downturns on vulnerable groups, and suggest that an export slowdown not only increases economically motivated crimes but also significantly raises other types of crimes, such as traffic offenses and violent crimes, as shown in Table 4.

5.3 Provision of Local Public Services

Another channel through which the adverse trade shocks may induce crime is by reducing local fiscal revenue. Shrunk fiscal income limits the provision of local public services (Feler and Senses, 2017), which, as a consequence, makes crime incidents harder to prevent. We explore the likelihood of this channel by examining two categories of government spending that are most relevant to social stability maintenance.[42] The first is the fiscal spending on public security, which includes all expenses by the People's Armed Police, judicial and prosecutorial system, and public security organs. The second is the "social spending", which is the sum of expenditure on public services, education, social security, medical services, and urban and rural affairs. As the budgets of China's government are usually initiated one year in advance, public expenditure is likely delayed in response to export shocks. For this reason, the export shock variables are measured at $t-1$.

The results are presented in Table 8, where we use the log change in fiscal revenue or

[42] There is no one-step repository of city-level fiscal data in China, thus we collect detailed fiscal expenditure information from several sources, including the Fiscal Statistical Yearbooks published by the provincial Bureaus of Finance, the Statistical Yearbooks published by the provincial Bureaus of Statistics and by the city-level Bureaus of Statistics, as well as the balance sheets from city government websites when there are missing values in publicly available sources.

Table 9: Mechanism: Export Shocks and Other Economic Outcomes

Dep. Var.:	△ Economic outcome$_{ct}$								
	(1) Mfg. empl. / worker	(2) Svc. empl. / worker	(3) Ind. output per capita	(4) Log # of new firms	(5) Log # of new Mfg. firms	(6) Log # of new Svc. firms	(7) Log # of new firms*	(8) Log # of new Mfg. firms*	(9) Log # of new Svc. firms*
ExpShock$_{ct}$	0.017**	-0.017	1.321***	0.453**	0.562***	0.545**	0.407**	0.489***	0.487**
	(0.008)	(0.011)	(0.435)	(0.184)	(0.213)	(0.219)	(0.167)	(0.188)	(0.199)
City FE	Y	Y	Y	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y	Y	Y	Y
First-stage F-stat	17.49	17.49	12.67	7.38	7.38	7.38	7.38	7.38	7.38
Observations	2,265	2,265	1,681	1,896	1,896	1,896	1,896	1,896	1,896

Notes: Observations are at the city-year level. Dependent variables in columns (1) to (3) are the change in manufacturing and service employment share, and the log change in industrial output per capita in year t respectively, which are computed from the China City Statistical Yearbooks (2011 - 2018). Dependent variables in columns (4) to (9) are the log change in the number of new firms in year t, which are computed from the China Administrative Firm Registration Database (2011 - 2017). In columns (4), (5), and (6), new firms are those created and assigned with (unique) administrative registration number in each year; in columns (7), (8), and (9), new firms refer to those whose administrative registration number was changed in each year (e.g., due to M&A), which we index with "*".The main explanatory variable $ExpShock_{ct}$ and the Bartik IV $ExpShock_{ct}^{IV}$ are constructed according to equations (1) and (3), respectively. All columns report IV estimates, which include city- and year-fixed effects. Robust standard errors are reported in parentheses and are clustered at the city level. *** p<0.01, ** p<0.05, * p<0.1.

expenditure per capita of city c between year t and $t-1$ as the dependent variables. Column (1) evaluates total fiscal revenue, column (2) assesses total fiscal expenditure, column (3) examines public security spending, column (4) considers social spending, and column (5) analyzes total spending on stability maintenance, which is the sum of public security and social spending. Columns (1) and (2) shows that the export slowdown significantly increases fiscal revenue and fiscal expenditure. Notably, column (4) reveals that cities experiencing a more severe export slowdown spend more on social services, but column (3) shows no significant increase in public security expenditure. Overall, column (5) demonstrates that total spending on stability maintenance increases in response to the export slowdown.

In summary, it appears unlikely that rising crime rates is due to reduced local fiscal revenue. Given the documented link between worsening labor market outcomes and higher crime rates, these results also suggest that rising spending on stability maintenance is a response to increased crime incidence. By allocating more resources to public services, cities attempt to mitigate the adverse impact of export shocks.

5.4 Other Economic Outcomes

Finally, we present evidence on how the export slowdown affects a broader set of economic outcomes, as shown in Table 9. Consistent with previous findings on employment and earnings, we find that the export slowdown leads to a significant decline in the share of manufacturing employment (column (1)) and has no significant effect on employment composition in the services sector (column (2)). Additionally, the adverse export shock results in a decrease in gross industrial output per capita (column (3)).

Columns (4) to (9) examine the patterns of firm entry across cities and sectors, focusing on the number of new firms based on their registration information from the China Administrative Firm Registration Database. In columns (4) to (6), new firms are defined as those created and assigned a unique administrative registration code each year. In columns (7) to (9), new firms are defined as those whose administrative registration code changes each year, indicated by an asterisk (*).[43] Across all regressions, we observe that the export slowdown discourages firm entry in both the manufacturing and services sectors.

These results further reinforce the finding that negative export shocks lead to worsening labor market conditions, which can negatively impact individuals' mental well-being. In Appendix F, we demonstrate that these empirical findings align with the predictions of a simple trade model featuring heterogeneous agents and an endogenous choice of occupation among criminals, workers, and entrepreneurs. This evidence supports the conventional wisdom on the determinants of crime, such as the Becker-type explanation (Becker, 1968; Britto et al., 2022; Bennett and Ouazad, 2020). A novel contribution of this study is the incorporation of psychological factors as drivers of criminal behavior.

6 Conclusion

In this paper, we investigate the criminogenic consequence of China's sharp export slowdown during a period of *secular stagnation* (Summers, 2014, 2015). Home to nearly 150 million manufacturing workers, China has been relying on exports to generate enough jobs in manufacturing and associated service sectors. Our study indicates a sizable impact of the export slowdown on crime rates: a reduction of $1,000 in a city's export per worker is predicted to induce 143 additional criminal cases per million population.

The analysis uses novel measures of various types of crime, which are constructed from detailed textual analysis of judicial documents disclosed by all levels of Chinese courts. Adopting the widely-used shift-share approach, we identify the causal effect of the export slowdown on crime rates using variations across cities in their exposure to exogenous foreign demand shocks. To the best of our knowledge, this is the first paper that exploits the detailed court judgment documents and establishes the causal linkage between export shock and crime in China.

Export slowdowns lead to substantial heterogeneity in the criminogenic response, depending on the industrial and demographic structure of cities. Cities with a higher reliance on manufacturing, greater proportions of young people, migrants, school dropouts, and higher

[43] In the second definition, a firm is considered a new entry if it is newly invested in by another firm, resulting in a new registration code.

initial crime rates—particularly those in the eastern region—experience larger increases in crime due to sluggish exports. We find that export slowdowns have more pronounced effects on offenses related to felony traffic violations, violence, prostitution, gambling, and resource appropriation activities such as robbery and stealing.

Our work demonstrates that adverse export shocks weaken labor market conditions, particularly for economically vulnerable individuals, leading to significant drops in jobs and wage earnings for the migrant population in urban areas. We also show that deteriorating psychological conditions are another driver of criminal behavior. More broadly, our research calls for greater attention to the socioeconomic consequences of economic slowdowns. As such, policymakers should prioritize policies that mitigate the disruptive impacts of stagnating exports, such as enhancing social safety nets, investing in mental health services, and creating targeted employment programs for affected groups. In a world where many advocate for de-globalization, it is crucial to develop strategies that support economic resilience and social stability amidst changing global trade dynamics.

7 Supplementary data

The data and codes for this paper are available on the Journal repository. They were checked for their ability to reproduce the results presented in the paper. The replication package for this paper is available at the following address: https://doi.org/10.5281/zenodo.15052806.

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